Improved Bounds on Optimal Exponents in Active Simple Hypothesis Testing.

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***Abstract*—The paper presents various methods for obtaining improved bounds on the optimal minimax error exponent in the Active Simple Hypothesis Testing (ASHT) problem with a given sample budget. The ASHT problem is a difficult problem studied by diverse communities like Information Theory, Bandits and Operations Research, but a clean characterization of this optimal exponent has not been available until now.**

This work combines a recent Partial Differential Equation (PDE) characterization of the optimal exponent (see [Vij25]) with stability bounds from the theory of viscosity solutions to Hamilton-Jacobi PDEs and convex analysis. This leads to bounds that depend on arbitrary convex functions, sets and lower Hamiltonian relaxations, a flexibility that immediately provides bounds that generalizes known results in the literature.

Note: THIS PAPER IS ELIGIBLE FOR THE STUDENT PAPER AWARD

***Index Terms*—Active Hypothesis Testing, Best Arm Identification, Fixed Budget, Partial Differential Equations.**

I. Introduction

In Active Simple Hypothesis Testing (ASHT), a controller interacts with the underlying unknown hypothesis through K actions and receives a sample observation from this hypothesis conditioned on this chosen action. It can adaptively choose its future actions based on the prior history of observations and actions. After the budget of T samples has been exhausted, it is required to output a hypothesis $\hat{i}_T \in \mathcal{H}$. The controller aims to minimize the worst-case error probability:

$$P_e(\mathcal{H}) := \max_{i \in \mathcal{H}} P_e^i(i \neq \hat{i}_T)$$

over the hypotheses $\mathcal{H}$. An optimal solution may be obtained through dynamic programming, but it is quite inefficient computationally and not very insightful. Instead, a natural goal is to study the optimal asymptotic ($T \to \infty$) exponent in the rate of decay of P_e. This leads us to studying the problem in a large deviations setting [Dem09].

This problem is motivated by many applications, including sensor management [Cas97], [AVF11], channel identification [Hay09] and planning in medical trials [Arm60]. This problem has been studied under various names of Channel Discrimination [Hay09], Controlled Sensing [NAV13], Ranking & Selection [GJ04] and Fixed-Budget Best Arm Identification [AB10], [KTH22] by the information theory, simulation and bandit research communities.

We provide tighter bounds on the optimal exponent of the error probability achievable by any feasible control strategy compared to prior works like [GJ04], [Hay09], [NAV13]. We do so by combining a characterization [Vij25] of the optimal exponent in terms of a Partial Differential Equation (PDE) and stability bounds [Abg94] for such Hamilton-Jacobi-Isaac (HJI) PDEs.

II. Problem Formulation

Consider a hypothesis testing scenario with m hypotheses, where the set of hypotheses is denoted by $\mathcal{H} = \{H_1, \cdots, H_m\}$. Each hypothesis is assumed to be simple, that is, it consists of a single channel. At each time step, the agent chooses an action a from the set $\mathcal{A}$ (with $|\mathcal{A}| = K$), and "nature" returns samples from the conditional distribution of action a according to the true underlying channel. We assume these conditional distributions have full support over the common finite set Y. The ith hypothesis is a channel whose input set is $\mathcal{A}$ -

$$H_i := \{P_a^i \mid a \in \mathcal{A}\}$$

for each $H_i \in \mathcal{H}$. The controller has a sample budget T, and the controller's policy π_T consists of T many *sampling rules*, denoted as π_T^t, for each $t \in [T]$ and a final *decision rule* $\hat{i}_T$. The sampling rule is *causal* and hence, each π_T^t is adapted to the sample filtration $\mathcal{F}_t$. The decision rule $\hat{i}_T \in [m]$ is adapted to the final sample filtration $\mathcal{F}_T$ and is interpreted as the final hypothesis choice of the controller. We denote the set of all such admissible causal controller policies for a horizon T as $\mathbf{\Pi}_T$. $\mathbf{\Pi}$ is the infinite sequence $(\pi_1, \ldots, \pi_T, \ldots)$ of such policies such that $\pi_T \in \mathbf{\Pi}_T, \forall T \in \mathbb{N}$.

Define the *error probability* of a policy π_T on the channel hypothesis $H_i \in \mathcal{H}$ to be $P_e(i, \pi_T) = P_{H_i, \pi_T}(\hat{i}_T = i)$.

Defintion 1. *The **error exponent** of an admissible sequence of controller policies $\Pi \in \mathbf{\Pi}$ for H_i, is defined as*

$$e(H_i, \Pi) = \liminf_{T \to \infty} -\frac{\log(P_e(i, \pi_T))}{T}$$

*and the **minimal** exponent over the hypotheses $\mathcal{H}$ as*

$$e_m(\mathcal{H}, \Pi) = \min_{H_i \in \mathcal{H}} e(H_i, \Pi).$$

*The **minimax-optimal** exponent is then defined as:*

$$e^*(\mathcal{H}) = \max_{\Pi \in \mathbf{\Pi}} \; e_m(\mathcal{H}, \Pi),$$

and a policy sequence Π^* *that attains this maxima*[1] *is called the* ***minimax-optimal policy***.

In this work, we show an alternate PDE based approach to upper and lower bounds for $e^*(\mathcal{H})$.

III. Related Work

[Hay09] studied the ASHT problem in the simplest case of $|\mathcal{H}| = 2$ (binary hypothesis testing) and showed that the non-adaptive strategy of playing the action that maximally discriminates the hypotheses in Chernoff Information is optimal.

[NAV13] extended this to the more general case of $|\mathcal{H}| > 2$. They give the upper bound

$$\min_{i \neq j} \max_a C(P_a^i, P_a^j)$$

for the optimal exponent, where $C(P, Q)$ denotes the Chernoff Information between P and Q. They also present an ASHT problem instance where adaptive strategies outperform open-loop/non-adaptive ones when $|\mathcal{H}| > 2$, in contrast to the case when $|\mathcal{H}| = 2$.

[KTH22] studied the problem of achieving and characterizing the minimax exponent in ASHT and used the data processing inequality to obtain a family of upper bounds indexed by $B \in \mathbb{N}$, which in the limit $B \to \infty$, converges to $e^*(\mathcal{H})$. They demonstrated that this exponent is achievable, although the policy proposed was intractable computationally. To side-step this difficulty, they construct a neural network-based algorithm that has good empirical performance but comes with no optimality guarantees.

[Vij25] showed that $e^*(\mathcal{H}) = V(0, 1)$, where the function V is the *viscosity solution* of the Hamilton-Jacobi-Isaacs (HJI) PDE:

$$\partial_t V + H(\nabla V) = 0 \tag{1}$$

with the initial condition $V(x, 0) = g(x) = x_{(2)}$ [2]. The Hamiltonian H has the form:

$$H(p) = \min_a H_a(p)$$

with $H_a(p) := \max_Q \left\{ -\sum_{i=1}^m p_i D\big(Q_a \,\|\, P_a^i\big) \right\}$[3] [4]. The HJI-PDE is a consequence of interpreting $e^*(\mathcal{H})$ as the value of a certain differential game between the controller and nature, wherein each tries to optimize the likelihood of error over all sample paths. Using this, [Vij25] showed upper bounds through a combined application of the comparison principle & the Hopf formula and obtained lower bounds through the algorithm based on Blackwell Approachability, which he proves leads to a good *subsolution* to HJI-PDE (1).

In this paper, we build on [Vij25]'s formulation and use techniques from viscosity solution theory of PDEs to obtain various bounds. Our bounds are achieved by either perturbing the initial condition g or the Hamiltonian H by convex and concave approximations. We obtain three families of bounds, one parametrized by any proper convex functions, the second by proper convex sets and the third by lower relaxations of the Hamiltonian. This improved flexibility generalizes prior bounds obtained in the literature (cf. [Hay09], [NAV13], [KTH22], [Vij25]).

[1] A dynamic programming argument shows the existence of Π^*.

[2] $x_{(i)}$ is the ith order statistic of the vector x.

[3] $D(.||.)$ is the KL divergence.

[4] Q_a is any probability measure on the common support Y.

IV. Preliminaries

Consider a Cauchy PDE with the following form

$$\phi_t + H(\nabla \phi) = 0$$

in $\mathbb{R}^m \times (0, +\infty)$ with an initial condition

$$\phi(x, 0) = \phi_1^0(x) + \phi_2^0(x), x \in \mathbb{R}^m$$

where H is a Lipschitz continuous function in $\mathbb{R}^m$ and ϕ_1^0, ϕ_2^0 are, respectively, convex and concave, uniformly Lipschitz functions. It is known that under these regularity conditions, the above Cauchy problem admits a unique Lipschitz continuous *viscosity solution*.

We aim to obtain upper and lower bounds on the viscosity solution and hence bounds on $e^*(\mathcal{H})$. To do this, we also need the *Hopf* and *Hopf-Lax* formula (see [Eva22]) that provides the exact solution when the initial condition ϕ^0 or the Hamiltonian is convex or concave. For the convex initial condition case,

$$\phi(x, t) = \sup_{v \in \mathbb{R}^m} [x \cdot v - \phi^{0*}(v) - tH(v)]$$

And for the concave case, we get

$$\phi(x, t) = \inf_{v \in \mathbb{R}^m} [x \cdot v - \phi^{0*}(-v) - tH(v)]$$

where $*$ represents the Legendre-Fenchel conjugate. A similar formula holds for the convex or concave Hamiltonian.

V. Upper and Lower Bounds

To obtain the Upper and Lower bounds we recall results from [Abg94]. However, there is a critical mistake in Abgrall's analysis, so we re-derive and present the corrected formula.

Let us begin by fixing notation. We denote by D_ϕ the set, for convex ϕ by

$$D_\phi = \{x \in \mathbb{R}^m, \phi^*(x) < +\infty\}$$

and for concave ϕ by

$$D_\phi = \{x \in \mathbb{R}^m, \phi^*(x) > -\infty\}$$

Theorem 1.

$$\inf_{v \in D_{\phi_2^0}} \sup_{w \in \mathbb{R}^m} \Big\{ x \cdot w - \phi_2^{0*}(v) - \phi_1^{0*}(v + w) - tH(w) \Big\}$$
$$\geq \phi(x, t) \geq$$
$$\sup_{v \in D_{\phi_1^0}} \inf_{w \in \mathbb{R}^m} \Big\{ x \cdot w - \phi_1^{0*}(v) - \phi_2^{0*}(v - w) - tH(w) \Big\}$$

Proof. We start by proving the lower bound. Since $\phi_1^0(x)$ is a convex function,

$$\phi_1^{0*}(v) = \sup_{x \in \mathbb{R}^m} [x \cdot v - \phi_1^0(x)]$$

$$\phi_1^{0*}(v) \geq x \cdot v - \phi_1^0(x)$$
$$\phi_1^0(x) \geq x \cdot v - \phi_1^{0*}(v)$$
$$\phi^0(x) \geq x \cdot v + \phi_2^0(x) - \phi_1^{0*}(v) := \xi_v(x)$$

That is, $\forall x, \phi^0(x) \geq \xi_v(x) = x \cdot v + \phi_2^0(x) - \phi_1^{0*}(v)$. Clearly, $\xi_v(x)$ is a concave function of x.

We would like to use the Comparison Principle for Viscosity Solutions [CL84] to compare the evolutions of the PDE with the initial conditions $\phi^0(x)$ and $\xi_v(x)$ respectively. Since $\xi_v(x)$ is a concave function, we can directly obtain its viscosity solution by the Hopf-Lax formula

$$\xi_v(x,t) = \inf_{w \in \mathbb{R}^m} \left\{ x \cdot w - \xi_v^*(-w) - tH(w) \right\}$$

Notice

$$\begin{aligned}
\xi_v^*(-w) &= \inf_{x \in \mathbb{R}^m} \left\{ + x \cdot w - \xi_v(x) \right\} \\
&= \inf_{x \in \mathbb{R}^m} \left\{ x \cdot (w - v) + \phi_1^{0*}(v) - \phi_2^0(x) \right\} \\
&= \inf_{x \in \mathbb{R}^m} \left\{ - x \cdot (v - w) + \phi_1^{0*}(v) - \phi_2^0(x) \right\}
\end{aligned}$$

Thus, we have

$$\xi_v^*(-w) = \begin{cases} +\infty, & v \notin D_{\phi_1^0} \\ \phi_2^{0*}(v - w) + \phi_1^{0*}(v) & v \in D_{\phi_1^0} \end{cases}$$

Putting this back inside the Hopf-Lax formula, we obtain, for $v \in D_{\phi_1^0}$ that

$$\xi_v(x,t) = \inf_{w \in \mathbb{R}^m} \left\{ x \cdot w - \phi_2^{0*}(v - w) - \phi_1^{0*}(v) - tH(w) \right\}$$

By the comparison principle we get, for any $v \in D_{\phi_1^0}$

$$\phi(x,t) \geq \xi_v(x,t) = \inf_{w \in \mathbb{R}^m} \left\{ x \cdot w - \phi_2^{0*}(v - w) - \phi_1^{0*}(v) - tH(w) \right\}$$

Supremizing over all feasible v we get

$$\phi(x,t) \geq \sup_{v \in D_{\phi_1^0}} \inf_{w \in \mathbb{R}^m} \left\{ x \cdot w - \phi_1^{0*}(v) - \phi_2^{0*}(v - w) - tH(w) \right\}$$

One obtains the upper bound using a similar argument. □

VI. Application to our PDE

A. Bounds using Abgrall's analysis

Now that we have proved this result, let us apply this to the HJI-PDE (1) To apply Theorem 1, we start by decomposing the initial condition $x_{(2)}$ as a sum of convex and concave functions.

Lemma 1.

$$x_{(2)} = \sigma_P(x) - \sigma_Q(x)$$

where σ_P, σ_Q represent the support functions[5] *of the convex bodies*

$$P = \text{convhull}(\{\mathbf{1} - e_i\}_i), \; Q = \text{convhull}(\{\mathbf{1} - e_i - e_j\}_{i,j})$$

respectively.

Proof. As

$$\begin{aligned}
x_{(2)} &= \left[\sum_i x_i - \min_i \{x_i\} \right] + \left[- \max_{|S| = n-2} \sum_{i \in S} x_i \right] \\
&= \sup_{\lambda \in P} \langle \lambda, x \rangle - \sup_{\mu \in Q} \langle \mu, x \rangle
\end{aligned}$$

Therefore,

$$x_{(2)} = \sigma_P(x) - \sigma_Q(x)$$

□

Fact 1. *For a convex body P, the Legendre-Fenchel conjugate of its support function satisfies*

$$\sigma_P^*(x) = \delta_P(x)$$

where $\delta_P(x) = 0$ for $x \in P$ and $+\infty$ otherwise.

See standard convex analysis texts like [Roc15], [HUL96] for a proof of this. Using this sum of convex and concave functions decomposition, one obtains the following upper and lower bounds:

Theorem 2.

$$\begin{aligned}
&\inf_{v \in Q} \sup_{u \in P} \left\{ x \cdot (u - v) - tH(u - v) \right\} \\
&\geq V(x,t) \geq \\
&\sup_{v \in P} \inf_{u \in Q} \left\{ x \cdot (v - u) - tH(v - u) \right\}
\end{aligned}$$

And, since, $e^*(\mathcal{H}) = V(0,1)$ satisfies

$$\begin{aligned}
&\inf_{v \in Q} \sup_{u \in P} \left\{ - H(u - v) \right\} \\
&\geq e^*(\mathcal{H}) \geq \\
&\sup_{v \in P} \inf_{u \in Q} \left\{ - H(v - u) \right\}
\end{aligned}$$

Proof. This is a direct application of Theorem 1. Use Fact 1 with the decomposition

$$x_{(2)} = \sigma_P(x) - \sigma_Q(x)$$

□

Using Theorem 1, one can get another class of bounds. This is done by noting that the sum of convex and concave function decomposition is not unique. Let $\mathscr{C}(\mathbb{R}^m)$ be the set of *all* convex functions on $\mathbb{R}^m$ and $\mathscr{C}_P(\mathbb{R}^m)(\subset \mathscr{C}(\mathbb{R}^m))$ be the subset of all proper, convex, and lower semi-continuous functions.

[5] $\sigma_R(x) := \sup_{y \in R} \langle y, x \rangle$, see for example [Roc97].

To begin, let $F \in \mathscr{C}_P(\mathbb{R}^m)$. Consider the initial function decomposition:

$$x_{(2)} = \tilde{\phi}_1^0 + \tilde{\phi}_2^0 = [\sigma_P(x) + F^*(x)] - [\sigma_Q(x) + F^*(x)]$$

where F^* is the Legendre-Fenchel conjugate of F.

Fact 2. *Let $f, g \in \mathscr{C}(\mathbb{R}^m)$ be proper, convex and lower semi-continuous. Then*

$$[f+g]^* = (f^* \,\square\, g^*),$$

where $\square$ denotes the infimal convolution.

Again, reader should look at standard convex analysis books VI-A for a proof.

Theorem 3. *Let F be a proper, convex and lower semi-continuous function. Then,*

$$\inf_{v \in D_{\tilde{\phi}_2^0}} \sup_{z \in \mathbb{R}^m} \Big\{ (\delta_Q \,\square\, F)(v) - (\delta_P \,\square\, F)(z) - H(z-v) \Big\}$$
$$\geq e^*(\mathcal{H}) \geq$$
$$\sup_{v \in D_{\tilde{\phi}_1^0}} \inf_{z \in \mathbb{R}^m} \Big\{ -(\delta_P \,\square\, F)(v) + (\delta_Q \,\square\, F)(z) - H(z-v) \Big\},$$

where $\square$ denotes the infimal convolution.

Proof. We have

$$x_{(2)} = [\sigma_P(x) + F^*(x)] - [\sigma_Q(x) + F^*(x)] = \tilde{\phi}_1^0 + \tilde{\phi}_2^0.$$

We will apply Theorem 1. To do this, we need $\tilde{\phi}_1^{0*}$ and $\tilde{\phi}_2^{0*}$. Note by Lemma 2,

$$\tilde{\phi}_1^{0*}(z) = [\sigma_P + F^*]^*(z) = (\delta_P \,\square\, F)(z)$$

and similarly

$$-\tilde{\phi}_2^{0*}(v) = (-\tilde{\phi}_2^0)^*(v) = [\sigma_Q + F^*]^*(v) = (\delta_Q \,\square\, F)(v).$$

Now, applying Theorem 1 one gets the result. $\square$

We also have the following result by optimizing the above over $F \in \mathscr{C}_P(\mathbb{R}^m)$:

Corollary 1.

$$\inf_{\substack{F \in \mathscr{C}_P(\mathbb{R}^m) \\ v \in D_{\tilde{\phi}_2^0}}} \sup_{z \in \mathbb{R}^m} \Big\{ (\delta_Q \,\square\, F)(v) - (\delta_P \,\square\, F)(z) - H(z-v) \Big\}$$
$$\geq e^*(\mathcal{H}) \geq$$
$$\sup_{\substack{F \in \mathscr{C}_P(\mathbb{R}^m) \\ v \in D_{\tilde{\phi}_1^0}}} \inf_{z \in \mathbb{R}^m} \Big\{ -(\delta_P \,\square\, F)(v) + (\delta_Q \,\square\, F)(z) - H(z-v) \Big\}$$

We will now derive the bounds for some specific functions $F \in \mathscr{C}_P(\mathbb{R}^m)$. We study the simple classes of support functions and squared Euclidean distances. The intuition for these bounds comes from the following. Consider the lower bound in Theorem 2. One wishes to make the supremizing player have more choices while lessening the choices of the infimizing player, and vice versa for the upper bound.

Corollary 2. *For R a convex body take $F = \sigma_R$,*

$$\inf_{v \in D_{\tilde{\phi}_2^0}} \sup_{z \in \mathbb{R}^m} \Big\{ (\delta_Q \,\square\, \sigma_R)(v) - (\delta_P \,\square\, \sigma_R)(z) - H(z-v) \Big\}$$
$$\geq e^*(\mathcal{H}) \geq$$
$$\sup_{v \in D_{\tilde{\phi}_1^0}} \inf_{z \in \mathbb{R}^m} \Big\{ -(\delta_P \,\square\, \sigma_R)(v) + (\delta_Q \,\square\, \sigma_R)(z) - H(z-v) \Big\}$$

This gives a bound based on arbitrary convex sets R.

Corollary 3. *For $F(x) = \|x-c\|_2^2$,*

$$\inf_{v \in D_{\tilde{\phi}_2^0}} \sup_{z \in \mathbb{R}^m} \Big\{ \operatorname{dist}(v,\, Q+c) - \operatorname{dist}(z,\; P+c) - H(z-v) \Big\}$$
$$\geq e^*(\mathcal{H}) \geq$$
$$\sup_{v \in D_{\tilde{\phi}_1^0}} \inf_{z \in \mathbb{R}^m} \Big\{ -\operatorname{dist}(v,\, P+c) + \operatorname{dist}(z,\; Q+c) - H(z-v) \Big\}$$

where $\operatorname{dist}(x, C)$ denotes the euclidean distance of x from a set C.

In Corollary 3, choosing the distance function penalizes the players from moving too far away from the sets P and Q, but the penalization is not as strong as $F \equiv 0$. The hope is to find a sweet spot.

One can write $-H(p) = \max_a(-H_a(p))$, switch $\inf, \max$ and obtain the exact infima using the fact that

$$\nabla_v \operatorname{dist}(v, Q+c) = 2(v - \pi_{Q+c}(v))$$

where $\pi_R(v) = \arg\min_{q \in R} \|v-q\|_2^2$.

Another natural choice for F is δ_R. However, one can easily prove that for that choice of F, the bound obtained would be necessarily worse than Theorem 2.

B. Bounds using a Hamiltonian relaxation

The last section was all about using Abgrall's analysis to obtain bounds on the behavior of the viscosity solution, and in particular, $e^*(\mathcal{H})$.

In this section, we will obtain bounds using a different approximation. Let $w = \{w_a\}_{a \in \mathcal{A}} \in \Delta_K$. Since $H_a(p)$ is convex, so is $\sum_{a \in A} w_a H_a(p) = \tilde{H}_w a(p)$.

We use this to relax the Hamiltonian and obtain a lower bound using the Hopf-Lax formula.

Theorem 4 (Hamiltonian Relaxation)**.**

$$V(x,t) \geq \sup_{w \in \Delta_K} \min_{y \in \mathbb{R}^m} \left\{ t\tilde{H}_w^* \left(\frac{x-y}{t} \right) + y_{(2)} \right\}$$

And, in particular,

$$e^*(\mathcal{H}) \geq \sup_{w \in \Delta_K} \min_{y \in \mathbb{R}^m} \left\{ \tilde{H}_w^*(-y) + y_{(2)} \right\}$$

Proof.

$$H(p) = \min_{a \in A} H_a(p) \leq \sum_{a \in A} w_a H_a(p) \tag{2}$$

where $w = \{w_a\}_{a \in \mathcal{A}} \in \Delta_K$. Since $H_a(p)$ is convex, so is $\sum_{a \in A} w_a H_a(p) = \tilde{H}_w(p)$. Consider the PDE (1) with the relaxed Hamiltonian,

$$\partial_t \tilde{V} + \tilde{H}_w(\nabla \tilde{V}) = 0 \tag{3}$$

and the same initial function. By the stability of viscosity solution [CL84] and Hopf-Lax, we get

$$V(x,t) \geq \tilde{V}(x,t) = \min_{y \in \mathbb{R}^m} \left\{ t\tilde{H}_w^* \left(\frac{x-y}{t} \right) + y_{(2)} \right\}$$

Optimizing over the allocation $w = \{w_a\}_{a \in \mathcal{A}}$ we get

$$V(x,t) \geq \sup_{w \in \Delta_K} \min_{y \in \mathbb{R}^m} \left\{ t\tilde{H}_w^* \left(\frac{x-y}{t} \right) + y_{(2)} \right\}$$

which proves the result.

□

VII. Conclusion.

We obtained flexible upper and lower bounds to the minimax-optimal error exponent $e^*(\mathcal{H})$, by utilizing the PDE characterization (1) along with comparison results for viscosity solutions on convex or concave approximations to Hamiltonian and initial conditions of the PDE. The bounds obtained are more general than prior work and are flexibly parametrized by convex functions, sets and lower Hamiltonian relaxations.

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